

Tech & Teach

Digital Chip Design Program

Design RTL

names:

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Task :0.2

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Task topic:

Data flow

Version 1.0

Task: 1.1

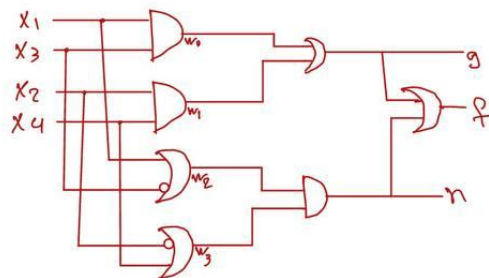
```
Project Summary x Schematic x DAY 2 TUESDAY second try.sv x
C:/Users/HUAWEI/Desktop/Vivado/DAY 2 TUESDAY second try/DAY 2 TUESDAY second try.srcs/sources_1/new/DAY 2 TUESDAY second try.sv
1 module lunch
2 # (parameter n = 4 )
3 {
4 input logic [n-1:0] X1,
5 input logic [n-1:0] X2,
6 input logic [n-1:0] X3,
7 output logic F
8 };
9 logic w0, w1;
10 assign w0 = X1 & X2 ;
11 assign w1 = ~X2 & X3 ;
12 assign F = w0 | w1 ;
13 endmodule
```

Task: 1.2

Task #1.2

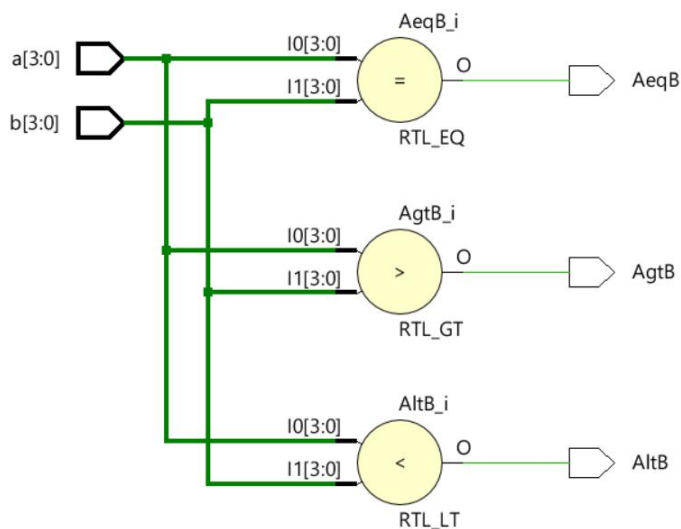
Given the following SystemVerilog code draw the microarchitecture:

```
1 module comb_circuit(f, g, h, x);
2
3 input [4:1] x;
4 output f, g, h;
5
6 wire [3:0] w;
7
8 assign w[0] = x[1] & x[3];
9 assign w[1] = x[2] & x[4];
10 assign w[2] = x[1] | ~x[3];
11 assign w[3] = ~x[2] | x[4];
12
13 assign g = w[0] | w[1];
14 assign h = w[2] & w[3];
15 assign f = g | h;
16
17 endmodule
```



Task: 1.3

```
1 module task2
2     #(parameter n = 4)
3     (
4     input logic [n-1:0]a,
5     input logic [n-1:0]b,
6     output logic AeqB,
7     output logic AltB,
8     output logic AgtB
9     );
10    assign AeqB =(a == b)? 1 : 0;
11    assign AltB =(a < b)? 1 : 0;
12    assign AgtB =(a > b)? 1 : 0;
13
14 endmodule
```



Work distribution

Section	shaher	Suliman	Abdullah
Report	✓		
Task 1.1			✓
Task 1.2	✓		
Task 1.3		✓	

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